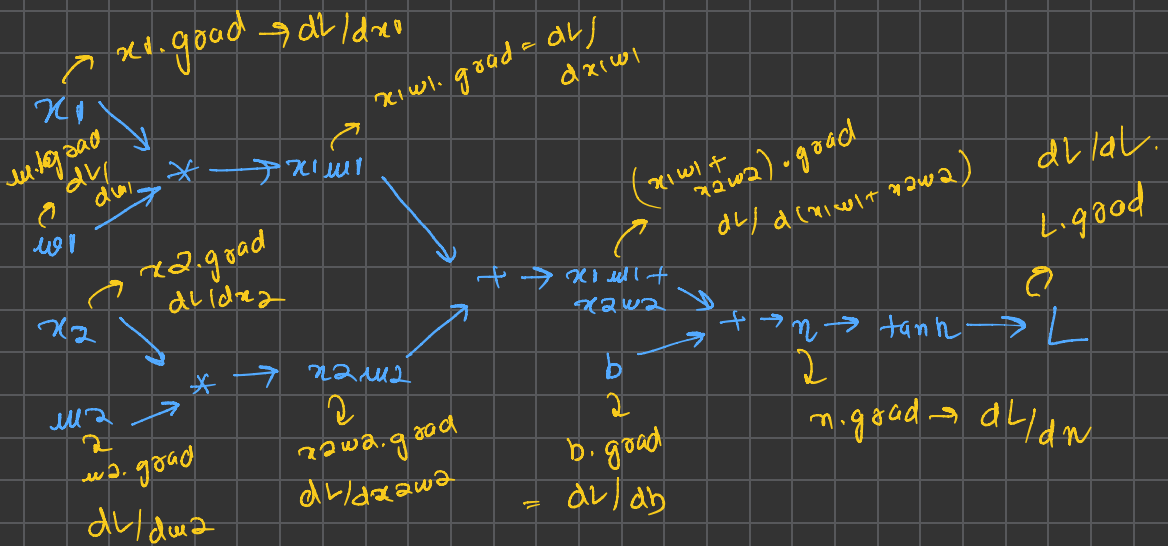
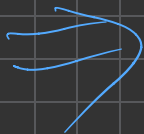


Topological sort for Neural Network



Important's

- 1) Neural Networks can never have, cycles in these computational graphs
- 2) So, these graphs are always DAG,
DAG $\rightarrow$ Directed Acyclic Graph




```

1 topo = []
2 visited = set()
3 def build_topo(v):
4     if v not in visited:
5         visited.add(v)
6         for child in v.prev:
7             build_topo(child)
8     topo.append(v)

```

Do of
run

```

topo = []
visited = set()
build_topo(L)
visited.add(L)
L.prev = [n]

```

```

build_topo(n)
visited.add(L, n)
n.prev = [(x1w1 + x2w2,
            b)]

```

```

build_topo(x1w1 + x2w2)
visited.add(L, n,
            x1w1 + x2w2)

```

```

x1w1 + x2w2.prev
= [x1w1, x2w2]

```

~~build_topo(x1w1)~~

visited.add(x, L, x1w1 + x2w2, x1w1)

x1w1.prev = [x1w1]

build_topo(x1)

visited. at $(L, n, (x1w1 + x2w2, x1w1, x1))$

topo. = $[x1]$

~~build topo (w1)~~

vis = $(L, n, (x1w1 + x2w2), x1w1, x1, w1)$

topo = $[x1, w1]$

topo.append = $[x1, w1, x1w1]$

~~build topo (x2w2)~~

vis = $(L, n, (x1w1 - x2w2), x1, w1, x1w1, x2w2)$

$x2w2. prev = [x2, w2]$

~~build topo (x2)~~

vis =

$(L, n, (x1w1 - x2w2), x1, w1, x1w1, x2w2, x2)$

$$t_{\text{ops}} = [x_1, w_1, x_1 w_1, x_2]$$

~~build_tops(x2)~~

$$vis = (x_1, w_1, x_1 w_1 + x_2 w_2, L, n, x_2, x_2 w_2)$$

$$t_{\text{ops}} = [x_1, w_1, x_1 w_1, x_2, w_2]$$

$$t_{\text{ops}} = [x_1, w_1, x_1 w_1, x_2, w_2, x_2 w_2]$$

$$t_{\text{ops}} = [x_1, w_1, x_1 w_1, x_2, w_2, x_2 w_2, x_1 w_1 + x_2 w_2]$$

~~build_tops(b)~~

$$vis = (x_1, w_1, x_1 w_1, x_2, w_2, x_2 w_2, (x_1 w_1 + x_2 w_2), L, n, b)$$

$$t_{\text{ops}} = [x_1, w_1, x_1 w_1, x_2, w_2, x_2 w_2, x_1 w_1 + x_2 w_2, b]$$

$$t_{\text{epo}} = [x_1, w_1, x_1 w_1, x_2, w_2, \\ x_2 w_2, x_1 w_1 + x_2 w_2, b, \\ n, L]$$

Final t_{epo} sort

